Research Proposal: Evaluating the Impact of Augmenting the VAEP Model with 360-Degree Freeze-Frame Spatial Context

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1. Introduction

The adoption of analytics in football has grown significantly in the last few decades (Goes et al., 2021; Olthof & Davis, 2025; Wakelam et al., 2022), enabling clubs to inform tactical decisions, coaching, and recruitment. Much innovation involves metrics that value player actions based on goal-scoring probabilities. One such metric is Valuing Actions by Estimating Probabilities (VAEP) (Decroos et al., 2019).

VAEP estimates the probability of scoring or conceding within the next k events to assign a value between -1 and 1 to each action. Like similar metrics, it typically uses action-centric features (e.g., start/end location, action type) but does not explicitly model wider spatial context (Sá-Freire, 2025). This spatial context - the position of teammates and opponents around the actor - determines a player's options, showing viable passing lanes or pressure from defenders and has been incorporated into other action value metrics successfully. Including spatial context in action valuation metrics like VAEP could provide utility in model predictive power and in tactical insight.

The StatsBomb Open Data repository (StatsBomb, 2026) provides event data alongside *three-sixty* freeze-frames , which capture the positions of nearby players. Such data can be used to incorporate spatial context as a predictive feature. **Figure 1** illustrates this: without spatial context, the pass appears to offer limited progression. Adding context reveals how opponents block the left side, making the pass into space on the right a valuable attacking opportunity (in this case, a long-ranged shot)[1]. Spatial context could enable finer-grained differentiation than aggregating all passes of a similar trajectory.

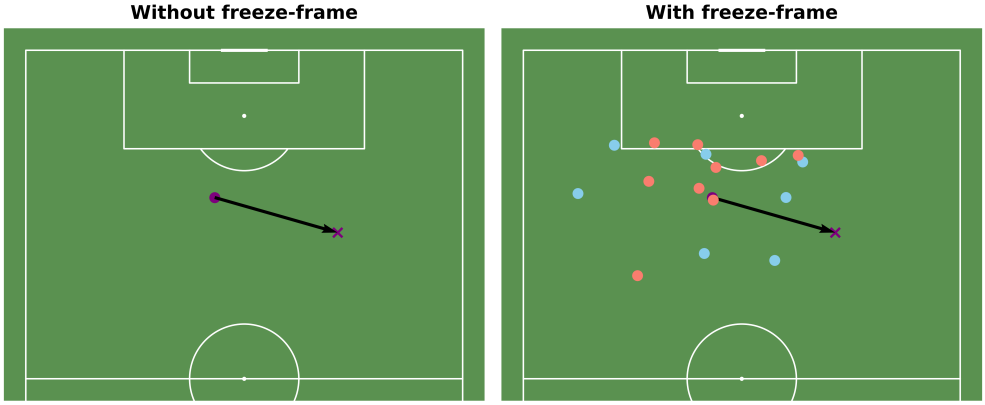


Figure 1: Pass Event With and Without Freeze-Frame

2. Research Objectives

This project aims to determine how adding spatial context features enhances the standard VAEP model. The objectives are:

- Develop representations of spatial context from StatsBomb 360 freeze-frame data using feature engineering and deep learning techniques.
- Train and evaluate VAEP models, both with and without the inclusion of spatial context representations, using supervised classification algorithms.
- Compare and interpret these VAEP models using appropriate performance metrics, measures of feature importance and an exploration of spatial context representation.

[1]The pass event id is '23b6b26a-293c-4d5e-9889-5a5ce0b7ad8c' and the subsequent long-ranged shot event id is 'd9d27d20-35bc-42bc-ac53-059ad519f781'

3. Background

Soccer analytics has developed metrics to value actions beyond simple outcomes like goals. Enhancing these models with player locations has improved performance and interpretability. This section introduces VAEP and shows how spatial context has been successfully incorporated into other metrics, motivating this project.

3.1. Valuing Actions by Estimating Probabilities (VAEP)

Decroos, Bransen, Van Haaren, & Davis (2019) define VAEP, which values an action a_i for team x by its change in the probability of scoring and conceding within the next k actions:

$$V(a_i, x) = \Delta P_{scores}(a_i, x) - \Delta P_{concedes}(a_i, x) \tag{1}$$

Each of these probability changes, $\Delta P_{scores}(a_i, x)$ and $\Delta P_{concedes}(a_i, x)$ are the difference between the game state following and preceding the action. Two supervised classifiers estimate these probabilities using a feature set based on the Soccer Player Action Description Language (SPADL) format (Decroos, Bransen, Van Haaren, & Davis, 2019), including action type and start/end locations, plus "complex features" (e.g., elapsed time) and "game context features" (e.g., scoreline).

The authors compared multiple classifier models, finding CatBoost to perform best across both the scoring and conceding models. They also used VAEP to generate player ratings (see Equation 2), finding that top-ranked players had higher total market valuations than those ranked by traditional goals+assists metrics, suggesting VAEP captures player value more comprehensively.

$$\text{rating} = \frac{90}{m} \sum_{a_i \in A_p^T} V(a_i) \tag{2}$$

One potential limitation of the VAEP model is that it lacks a representation of wider spatial context from the positions of teammates and opponents during actions (Sá-Freire, 2025), due to its use of event data only. Every action taken by a player exists within a spatial context which determines the options available to a player, influencing their decision making. Despite its addition having been successfully applied to a number of alternative action value metrics, spatial contexti in VAEP is understudied.

3.2. Applications of Spatial Context to Action Value Metrics

3.2.1. *Expected Goals (xG)*

xG values a shot as its probability of resulting in a goal, estimated using features like distance to goal and match context (e.g. set piece, open play).

Cefis & Carpita (2025) incorporate spatial context by counting the number of defenders positioned ahead of the shooter (higher x-coordinate than the shot location) within a fixed lateral band ($\pm 10\%$ in y). From this, they construct two binary variables, `D1.OpponentsPlayer` and `D2.OpponentsPlayer`, indicating whether there is at least one or more than one defender in this region, respectively. Furthermore, `GK_x_coordinate` and `GK_y_coordinate` represented the opponent goalkeeper location. Their results show that these spatial context features (except `GK_y_coordinate`) had negative coefficients in their logistic regression xG model, indicating reduced log-odds of scoring due to a unit increase in these features, respectively. This effect is strongest for `D2.OpponentsPlayer` (coef -6.13 to -5.69), reflecting a substantial reduction in scoring probability when multiple defenders are nearby. These spatial features, along with matching intuitions regarding opposition proximity, also contribute to their model outperforming the *understat* xG baseline in terms of precision and F1-score.

3.2.2. *Expected Threat (xT)*

xT values actions based on where the ball moves from and to by modelling transitions uses a Markov process on a discretised pitch (Singh, 2018). The value of a zone is defined iteratively, combining the immediate shooting threat and the potential future threat from passing or carrying.

Everett et al. (2022) introduce spatial context to xT's transition matrix $T_{(x,y)}$ via a pitch control (Spearman et al., 2017) and interception model (Burriel & Buldú, 2021) which was used to train a convolutional neural network (CNN) to produce the transition probabilities for their contextual transition matrix $T^{\star}_{(x,y)}$. The pitch control model considers the probability of a player controlling the ball as a function of their distance from opposition players, whilst the interception model introduces an additional transition probability, that being the probability of the ball being intercepted when moving from zone (x, y) to its intended destination (z, w). The authors found introducing spatial context improved model fit, reducing the log likelihood of the xT model from -5.137 to -4.151. They also found useful applications in tactical analysis, such as enabling the authors to apply a Basin-Hopping optimiser (Wales & Doye, 1997) to find more optimal defensive shapes than which occurred during actions, using their derived Threat Above Expected (TAx) metric as the algorithm's loss function.

3.2.3. *Expected Possession Value (EPV)*

Expected possession value (EPV) (Fernandez et al., 2021) is an action valuation metric, adapted from sports like basketball (Cervone et al., 2016), that uses tracking data to assign value to a game state at time t by considering all possible subsequent actions.

$$\mathrm{EPV}_t = \sum_{a \in A} \mathbb{E}[G \mid A = a, T_t] \quad \mathbb{P}(A = a \mid T_t) \tag{3}$$

Equation 3 defines EPV at time t for an action as the expected value of G (e.g., goal scored) given action a in spatiotemporal context T_t, multiplied by the probability of a occurring in T_t. Summing over all a in action set A yields the EPV of the game state.

T_t reflects EPV's dependence on spatial context. Fernandez, Bornn, & Cervone (2021) list features like pitch control, number of players within 3m of the ball, and number within the

area bounded by the ball and the opponent's goalposts. The authors illustrate EPV's utility by analysing Liverpool's 2014/15 on-ball (passes with positive EPV) and off-ball (pitch locations with positive EPV if a pass arrived) advantage against different formations. This reveals where Liverpool succeeded or failed in passing decisions (on-ball), and which pitch regions to target during build-up (off-ball). Such analysis aids tactical decision-making: teams can evaluate, against opposition pressing structures, where to progress the ball for maximum effectiveness in build-up play, and assess their current approach.

3.2.4. *Summary*

The discussed applications demonstrate the utility provided by incorporating spatial context into action valuation models. Given the lack of investigation into augmenting VAEP models with wider spatial context, this project aims to address this research gap and explore what role representations of spatial context play in altering model behaviour and performance.

4. Methodology

4.1. Datasets

This project will use data from StatsBomb's Open Data repository (StatsBomb, 2026). This project will use *events* and *three-sixty* data corresponding to 292 matches from UEFA Men's Euro 2020 & 2024, UEFA Women's Euro 2022 & 2025, FIFA Men's World Cup 2022, FIFA Women's World Cup 2023. Unlike most competitions in this repository, *three-sixty* data is available for every match in this sample, enabling player positional context to be incorporated into feature engineering for 1,052,116 match events.

Atomic-SPADL (Davis et al., 2020) will be used in this project for engineering the baseline feature set. Atomic-SPADL represents an action without considering its result (success vs failure), unlike SPADL which includes the result. An example of the effect this has on action representation is that a pass (one action under SPADL) is separated into two actions: the giving of the pass, and the reception of the pass. An additional "interception" action type is introduced to enable incomplete passes to be captured. This atomic-SPADL format will then be augmented in this project by the addition of engineered spatial features.

4.1.1. *Limitations of Datasets*

Football analysts often use tracking data (Goes et al., 2021), providing continuous records of player/ball positions rather than discrete time events. This allows modelling match actions as continuous processes, accounting for whole-team positioning and movement unlike events data. However, such tracking data is often proprietary and unavailable for public use (Bassek et al., 2025), limiting this project to discrete events and losing informative context (e.g. player movement direction and velocity). This limits the scope of the present project to viewing events as discrete in time, thus losing some surrounding context which may be informative (e.g. player movement direction and velocity). Events data is nonetheless relatively comprehensive, covering far more matches than public tracking data (Olthof & Davis, 2025). The use of StatsBomb's *three-sixty* data partially mitigates this by providing snapshots of proximal player positions at each event, permitting some inference on team positioning. But these snapshots only capture players within the broadcast camera's view (StatsBomb, 2026).

A further limitation concerns SPADL-formatted data. Recent work by Jo et al. (2026) shows some events data contains logical inconsistencies which SPADL does not correct. The authors propose the Verified Event Data Format for Reliable Soccer Analytics (VERSA), using a state-transition model to identify and fix inconsistencies, improving VAEP model training. Yet their results

regarding StatsBomb data indicate that atomic-SPADL still slightly outperforms VERSA. This result, together with the increased action-representation precision noted earlier, motivates using atomic-SPADL in this project. However, the potential data integrity issues highlighted by the authors will not be explicitly handled as implementing VERSA is non-trivial and thus beyond this project's scope.

4.2. Data Preprocessing

With StatsBomb events and *three-sixty* data collected, VAEP requires this data be converted into the Atomic-SPADL format. This preprocessing step can be achieved by the `socceraction` package (Socceraction Developers, n.d., Decroos, Bransen, Van Haaren, & Davis (2019)), which has functions such as `convert_to_atomic` and the `AtomicVAEP` class to handle StatsBomb events data for VAEP modelling.

This project intends to augment VAEP with additional spatial context features, outlined in Section 4.3 and Section 4.4, which deviates from the standard atomic-SPADL format, and requires manual data engineering and transformer model training.

4.3. Manual Spatial Feature Engineering

Previous research has represented spatial context using features engineered from spatial data. As described in Section 3.2.1, Cefis & Carpita (2025) derived two binary features from player x and y co-coordinates, indicating the number of players within the immediate frontal proximity of the shooter. Amichay et al. (2025) measured the areas of convex hulls and inner layers to represent team shapes. A convex hull, in this context, is the area inscribed between the outer-most positioned players on the pitch for a given team. A larger area corresponds to a wider/longer team shape, whereas a smaller area is narrower/shorter. Since this convex hull only considers the outer-most players, a second convex layer can be created from the players who do not contribute to the hull. The ratio of these two areas forms the author's derived Layer Ratio (LR) metric, which summarises the compactness of the team shape.

```
{
    "teammate": boolean,
    "actor": boolean,
    "keeper": boolean,
    "location": [float, float]
}
```

Figure 2: StatsBomb *three-sixty* freeze-frame JSON schema

Figure 2 outlines the JSON schema for a single player in a StatsBomb *three-sixty* freeze-frame. Each freeze-frame has an associated `event_uuid` and `visible_area` polygon which indicates the viewing area from the broadcast camera at the moment the action took place.

Thus, for this project, the following spatial features will be engineered for teammates and opponents, respectively, using *three-sixty* data. These are both inspired by the literature and chosen by myself:

- Actor's Euclidean distance to nearest teammate/opponent (meters)
- Number of teammates/opponents within x meters of the actor[2]
- Average and standard deviation of Actor's Euclidean distance to teammates/opponents in freeze-frame (meters)
- Teammate/opponent convex hull area (based on players within freeze-frame) (m^2)

[2] x will be left as a hyperparameter to be chosen during feature engineering and training, but will be reported as an exact value in the write up.

4.4. Spatial Embeddings: Set Transformer

Spatial embeddings are real-valued vectors representing geometric objects in a space, such as player co-ordinates.

As Figure 2 shows, StatsBomb *three-sixty* data doesn't label individual players in each freeze-frame. So, to create spatial embeddings, we need a *permutation-invariant* input layer, that is a layer which does not require features to enter in a particular order. A set transformer (Lee et al., 2019) fits this task. Like standard transformers (Vaswani et al., 2017), it uses multihead attention in encoder-decoder sublayers to learn relationships within a set. Permutation invariance is achieved by dropping position encodings and pooling outputs via pooling-by-multihead attention (PMA). PMA uses multihead attention to pool each set element's representation into k vectors, each summarising the whole set. k is usually 1, but $k > 1$ is allowed. The queries Q are learned seed vectors S. Re-ordering the set thus yields the same pooled representation, hence permutation invariance. Set transformers have been used successfully for player/ball trajectory prediction (Capellera et al., 2024; Kim et al., 2023). Though this domain differs from this present project, those studies confirm the method's suitability for encoding player positions.

I will use a set transformer to produce separate fixed-size embeddings for each set of players in each freeze-frame, using each key from Figure 2 (and separate out the x and y co-ordinates) as the feature set for each player in frame. To enable the set transformer to learn non-linear relationships, the feature set will first be projected into a higher-dimensional embedding space via a multilayer perceptron. These embeddings will thus form a feature set which will be compared against `baseline` and `baseline + manual features` VAEP models.

4.5. Model Training

This project will replicate the implementation of the same 4 classification techniques used by Decroos, Bransen, Van Haaren, & Davis (2019): logistic regression (Cox, 1958), random forest (Breiman, 2001), XGBoost (Chen & Guestrin, 2016) and CatBoost (Prokhorenkova et al., 2019). An ablation design will be used to compare the addition of manual spatial features, spatial embeddings and both simultaneously to the baseline VAEP model. This will result in 32 individual classification models.

For logistic regression, Decroos, Bransen, Van Haaren, & Davis (2019) report the use of L2 Regularisation and Limited-memory BFGS as the numeric solver. For their random forest models, they used 100 trees and 40 parallel threads. Regarding XGBoost, they again use 100 trees with a maximum depth of 3 nodes an the learning rate is 0.1 Lastly, the authors retain the default hyperparameters for CatBoost (CatBoost Developers, n.d.). Each of these hyperparameters (except the parallel threads) will be held constant in this project, and default hyperparameters will be used where left unspecified by the authors. Logistic regression, XGBoost and CatBoost each use cross-entropy loss as defaults, and random forest uses the Gini impurity split criterion.

It is expected, in line with Decroos, Bransen, Van Haaren, & Davis (2019), that CatBoost should perform best in terms of Brier score and AUC-ROC, within each model variant. Should deviations from the original VAEP results emerge, they will be discussed in the write up.

4.6. Model Evaluation Metrics

Decroos, Bransen, Van Haaren, & Davis (2019) used AUC-ROC and the Brier score (Brier, 1950) to evaluate the probability estimates of their scoring and conceding classifiers. This project will also add cross-entropy loss and expected calibration error (ECE) (Naeini et al., 2015) as metrics used to evaluate model performance. Cross-entropy loss penalises larger errors more harshly, and so is useful to assess the extent to which classifiers make predictions which deviate greatly.

$$\text{Log Loss} = -[y \log(p) + (1 - y) \log(1 - p)] \tag{4}$$

For example, if the model's estimated probability of a goal scored/conceded is 0.5 when the action does result in a goal within k events (i.e. $y = 1$), the loss would be $-[1 \times \log(0.5) + (1 - 1) \log(1 - 0.5)] \approx 0.693$, but if the estimated probability were 0.1, this would be ≈ 2.303.

ECE provides an indication of how well-calibrated predictions are by binning probability predictions and comparing the proportion of positive predictions in each bin to the average probability of that bin (e.g. 0.5–0.6 $\implies$ 0.55). A well-calibrated model should produce small deviations.

$$\text{ECE} = \sum_{i=1}^{K} P(i) \cdot |o_i - e_i| \tag{5}$$

4.7. Model Interpretability

4.7.1. *Permutation Importance*

Permutation importance estimates feature importance by randomly permuting one or more features and evaluating the change in model performance metrics. Features which, when permuted, result in a larger drop in model performance suggest greater importance. Care should be taken with multicollinearity, as permutations will not entirely disconnect their relationship with others model features since those relationships can continue to exist via their colinear counterparts. Permutation importance will thus be used in this project to highlight which features or set of features are most determinative of the VAEP values assigned to actions. The potential presence of multicollinearity could be addressed using hierarchical clustering (scikit-learn Developers, n.d.).

4.7.2. *Partial Dependence Plots (PDPs)*

Partial dependence plots (PDPs) (Friedman, 2001) are used to visualise the relationship between some set of features (usually just 1 or 2) and the predictions made by a machine learning model, holding other features constant. In this case, we can use PDP plots to explore the interpretable feature set, that is the baseline and manually-engineered features, as opposed to the spatial embeddings whose dimensions are not directly interpretable.

4.7.3. *k-Means Clustering*

We can also explore the spatial embeddings via k-means clustering, similarly to how DeepCluster applies k-means to image embeddings generated by a CNN (Caron et al., 2019). The benefit of using k-means clustering is that underlying relationships represented by spatial embeddings can be explored by comparing the properties of clusters. For instance, player position heatmaps for each cluster could be plotted to potentially reveal notable team shapes, or if certain clusters tend to involve certain action types over others. Both the trade-off between increasing k and model inertia (i.e. the "elbow method") and average silhouette score (Rousseeuw, 1987) will be used to derive k.

4.7.4. *Uniform Manifold Approximation and Projection (UMAP)*

To interpret the spatial embeddings, uniform manifold approximation and projection (UMAP) (McInnes et al., 2020) can be used as a dimensionality reduction technique to represent embeddings in a 2-or-3-dimensional space, which can then be visualised. The colour of data points within a UMAP can then be determined by properties such as the VAEP value, scoring/conceding probability or k-means clusters.

5. Project Planning

5.1. Estimated Timeline

This project will aim to adhere to the following estimated timeline, displayed via a Gantt chart. Given that the data being used in This project has already been collected, this timeline will omit data collection and initial preprocessing. This data collection process will be described in full in the write up. Furthermore, as of the time of producing this research proposal, as date for the oral examination has not been provided and as such time has not yet been allocated for preparation for this exam.

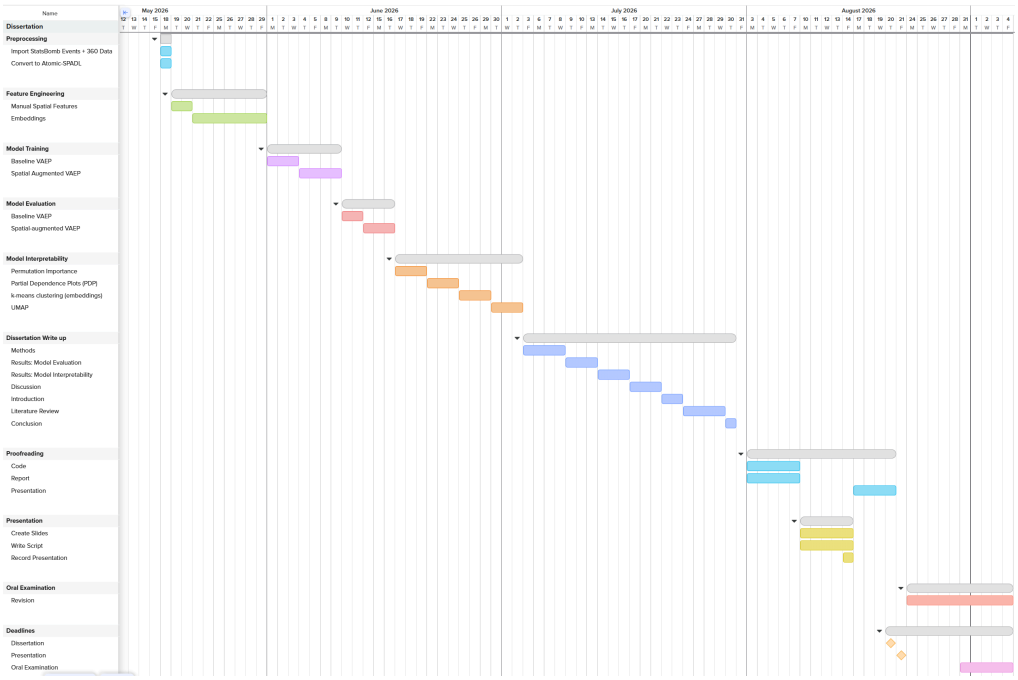


Figure 3: Project Gantt Chart

5.2. Prerequisite Resources

The following table lists the resources this project plans to use in exploring its research objectives.

Item	Value
OS	Windows 11
CPU	AMD Ryzen 5 3600 6-Core Processor (3.6 GHz)
RAM	16GB
GPU	Nvidia GeForce RTX 3070 8GB VRAM
Programming Language	Python 3.12.10
Libraries	pandas, numpy, scipy, matplotlib, seaborn, scikit-learn, pytorch, statsmodels, statsbombpy, socceraction, mplsoccer
Technical Publishing System	Quarto 1.6.42
IDE	Positron 2026.04.1

6. Conclusion

Soccer has seen huge growth in data science over the last couple of decades, largely driven by metrics like expected goals (xG), expected threat (xT) and expected possession value (EPV), each of which successfully incorporate spatial context. Valuing Action by Estimating Probabilities (VAEP) traditionally has not. This project therefore aims to develop representations of spatial context via manually engineered features and embedding representations using set transformers to then evaluate the impact of adding spatial context on VAEP model performance and behaviour.

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